Decarbonization Pathways for the Korean Petrochemical Industry: Model Assumptions and Results

PLANiT Institute

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Abstract

This report presents the key assumptions and results of a marginal abatement cost curve (MACC) analysis for decarbonizing Korea's petrochemical industry. Using a facility-level database of 248 plants with 100,066 kt/year capacity, we evaluate two primary pathways: NCC-Electricity and NCC-Hydrogen. Our analysis finds that net-zero emissions by 2050 is technically achievable through electrification, requiring $37.3 billion in cumulative investment and 230 TWh of annual renewable electricity.

1 Introduction

Korea's petrochemical sector produces approximately 46 $MtCO_2$/year at current operating rates (70%), making it one of the largest industrial emission sources. This report documents the data sources, model assumptions, and scenario results for achieving net-zero emissions by 2050.

2 Model Assumptions

2.1 Facility Database

Table 1: Facility Database Summary

Region	Facilities	Capacity (kt/yr)
Yeosu	87	37,216
Daesan	57	27,424
Ulsan	71	24,105
Onsan	14	7,331
Other	19	3,990
Total	**248**	**100,066**

2.2 Technology Parameters

Table 2: Technology CAPEX Assumptions (USD/t-C_2H_4/yr)

Technology	2025	2030	2040	2050	Source
NCC-Electricity	1,500	1,350	1,050	900	Toribio-Ramirez 2025
NCC-H2	1,700	1,300	935	780	Thunder Said Energy 2023
Heat Pump	800	640	480	400	McKinsey 2024
RDH	900	720	540	450	Coolbrook 2024

2.3 Energy Intensity

Table 3: Process Energy Intensity

Process	Parameter	Value	Source
Naphtha Cracker	Electricity	5.0 MWh/t-C_2H_4	BASF/SABIC/Linde 2024
Naphtha Cracker	H2 Consumption	0.2 t-H_2/t-C_2H_4	Lummus Tech 2023
Heat Pump	COP	4.0	Kosmadakis 2020
RDH	Efficiency	93%	Coolbrook 2024

2.4 Emission Factors

Table 4: Combustion Emission Factors

Fuel	tCO_2/GJ	Source
Naphtha	0.0542	IPCC 2019
LNG	0.0561	IPCC 2019
Fuel Gas	0.050	API Compendium 2021
Byproduct Gas	0.048	API Compendium 2021

2.5 Price Trajectories

Table 5: Energy Price Trajectories

Parameter	2025	2030	2035	2040	2045	2050
RE Price ($/MWh)	129	157	191	191	191	191
H2 Price ($/kg)	6.73	5.88	5.02	4.16	3.31	2.63

Source: PLANiT Institute (2025)

2.6 LCOH Calculation Assumptions

The Levelized Cost of Hydrogen (LCOH) is calculated using:

$$LCOH = \frac{CAPEX \times CRF + OPEX}{H_2\ Production} + Electricity\ Cost \tag{1}$$

Key parameters:

- Electrolyzer efficiency: 70% (HHV)
- Capacity factor: 90%
- Discount rate: 8%
- Electrolyzer CAPEX: $1,200/kW (2025) → $300/kW (2050)
- Stack lifetime: 10 years

3 Results

3.1 Baseline Emissions

At 70% operating rate, baseline emissions in 2025 are 46.3 $MtCO_2$/year. Business-as-usual projections reach 53.3 $MtCO_2$ by 2050.

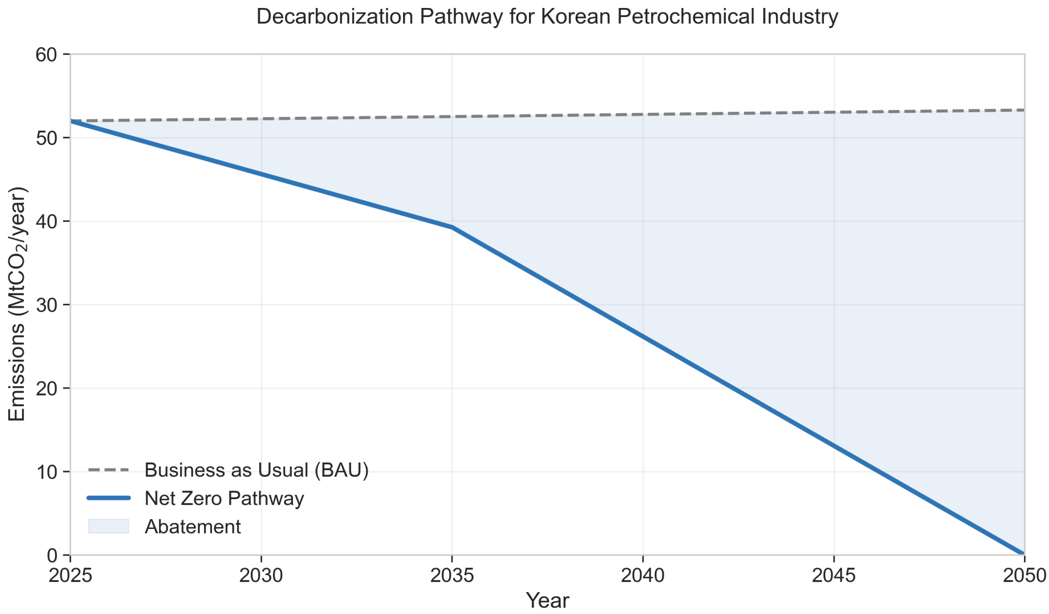

Figure 1: Decarbonization Pathway vs BAU (2025-2050)

3.2 Technology Contribution

The analysis assumes that the "NCC-Electricity" technology applies to the entire Naphtha Cracker process, decarbonizing all co-products (Ethylene, Propylene, Butadiene, etc.) simultaneously. This is a critical assumption as electrification replaces the fuel gas combustion in the cracking furnaces, which is the primary emission source for the entire facility.

Table 6: Technology Contribution to Net Zero (2050)

Technology	Abatement (Mt)	Share (%)	MAC ($/t$CO_2$)
NCC-Electricity	47.1	88.3	496
Heat Pump	3.6	6.7	301
RDH	2.6	4.9	1,086
Total	53.3	100.0	–

3.3 Scenario Comparison: NCC-Electricity vs NCC-H2

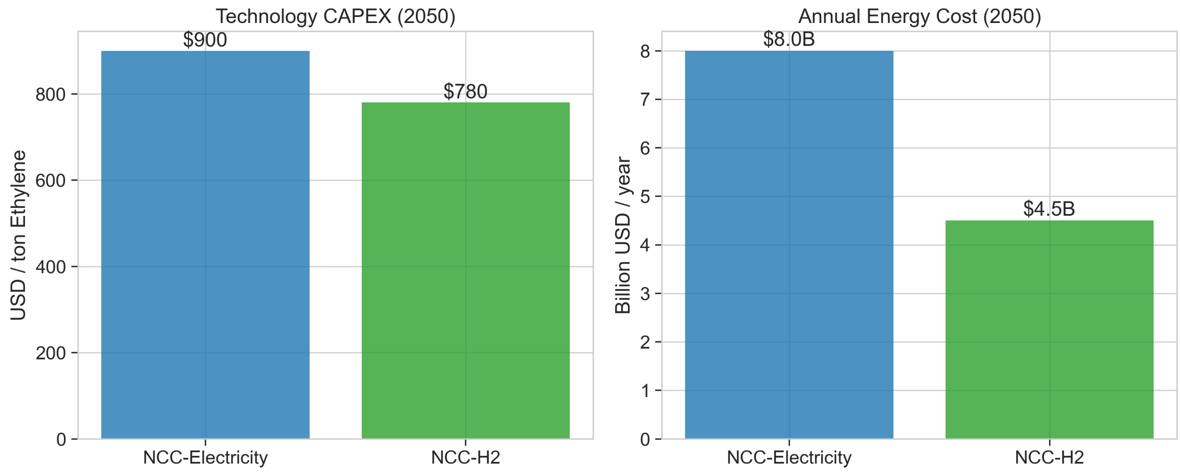

Figure 2: Scenario Comparison: CAPEX and Energy Cost in 2050

Table 7: NCC-Electricity vs NCC-H2 (2050)

Metric	NCC-Electricity	NCC-H2
Energy Consumption	42 TWh	1.7 Mt H_2
Energy Price (2050)	$191/MWh	$2.63/kg
Annual Energy Cost	$8.0B	$4.5B
CAPEX	$900/t-$C_2H_4$	$780/t-$C_2H_4$
Infrastructure	Grid upgrade	H_2 pipeline

The NCC-H2 pathway shows lower annual energy costs by 2050 due to declining hydrogen prices, but requires significant hydrogen infrastructure investment not captured in this analysis.

3.4 Regional Investment Requirements

Table 8: Regional Investment Summary

Region	Facilities	Emissions (Mt)	Investment ($B)	Elec (TWh)
Yeosu	87	18.5	13.8	85
Daesan	57	15.7	11.7	72
Ulsan	71	7.3	7.1	44
Onsan	14	4.6	4.7	30
Total	248	46.1	37.3	231

3.5 Cumulative Investment

Total investment over 2025-2050: \$37.3 billion.

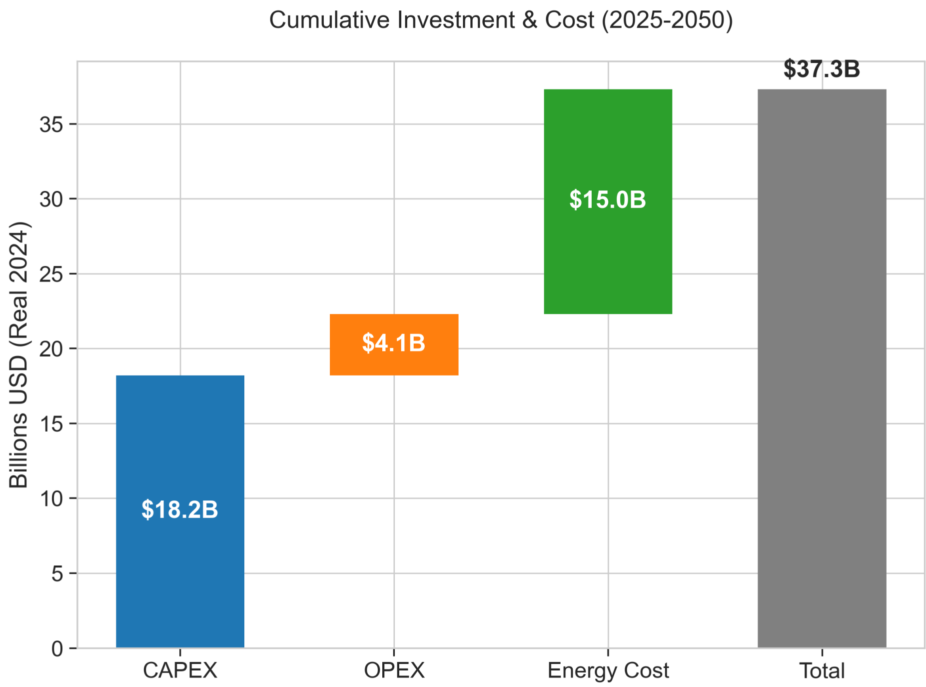

Figure 3: Cumulative Investment Breakdown (2025-2050)

4 Conclusion

Net-zero emissions for Korea's petrochemical industry is achievable by 2050 through:

1. Electrification of naphtha crackers (88% of abatement)
2. Heat pumps for low-temperature processes (7%)
3. RotoDynamic Heaters for high-temperature aromatics (5%)

The primary bottleneck is renewable electricity availability (231 TWh/year by 2050). The NCC-H2 pathway offers lower long-term energy costs but requires substantial hydrogen infrastructure.

5 References

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